

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 2 Outline

▶ IP address

- Allocation & Routing
- Schemes and classes
- Limitations to Classful Addressing
- Subnet Mask & VLSM
- Class exercise 4

▶ Subnetting

- Subnetting, Subnet & Subnet Mask
- VLSM & CIDR
- Calculating Subnets
- Class exercise 5
- Routing Table run-down

▶ ARP

- Frame & working
- Tables
- Class exercise 6

Recap

OSI 7 layer model	TCP/IP 5 layer model	Protocols	Protocol Data Unit (PDU)	Layer addressing
Application layer	Application layer	HTTP, FTP, SMTP, IMAP, DNS, Telnet, SNMP, ...	Data	Data
Presentation layer				
Session layer				
Transport layer	Transport layer	TCP, UDP, ...	Segment (TCP), Datagram (UDP)	Port address (Source, Destination)
Network layer	Internet layer	IP, ...	Packets	IP address (Source, Destination)
Data link layer	Network access layer	Ethernet, Wi-fi, Bluetooth, 3G, LTE, ...	Frames	MAC address (Source, Destination)
Physical layer	Physical layer	Copper, Fiber optic cables, Wireless transmitters	Bits	n/a

Open System Interconnection (OSI):
ITU-T standard X.200 (ISO/IEC 7498-1)

DoD Model: Currently
Maintained by IETF

IP address & their allocation

- ▶ IPv4: 4 bytes/octets (32 bits in total)
 - e.g., 163.1.125.98
 - In theory there are about 4 billion (2^{32}) available
- ▶ Controlled centrally by ICANN (The Internet Corporation for Assigned Names and Numbers)
 - A non-profit organization
 - Govern fairly strict rules on further delegation to avoid wastage
 - Have to demonstrate actual need for them!

Routing

- ▶ How does a device know where to send a packet?
 - Devices need to know what IP addresses are on directly attached networks
 - If the destination is on a local network, send it directly there
- ▶ If the destination address is not local
 - Most non-router devices just send everything to a single local router (default gateway)
 - Routers need to know which network corresponds to which IP address (range)

IP Address Scheme Categories

- ▶ Conventional (i.e., Classful) Addressing
 - Dividing line (b/w network & host bits) occurs only at octet boundaries
 - Class A, B, and C depending on how many octets for network ID and host ID
- ▶ Subnet (i.e., Classless) Addressing
 - 3-tier system:
network ID (NID), subnet ID (SID), host ID (HID)
- ▶ E.g.,
 - 24 (NID) + 8 (HID) $\rightarrow$ Classful (Class C)
 - 24(NID) + 3(SID) + 5 (HID) $\rightarrow$ Classless

Classful IP Address Scheme

- ▶ IP addresses are divided into 5 classes, each of which is designated with the alphabetic letters A to E
- ▶ Class D addresses are used for multicasting
 - ▶ Not assigned to individual devices like regular IPs. Instead, devices subscribe to multicast groups
- ▶ Class E addresses are reserved for testing and research purposes
 - ▶ Most systems will not route or allow Class E traffic unless specifically configured

Classful IP Address Scheme: Classes (Cont.)

- ▶ The 5 IP classes are split up based on the value in the 1st octet:

Class	First Octet Value	Usage
Class A	0 ~ 127	Unicast, Large networks
Class B	128 ~ 191	Medium networks
Class C	192 ~ 223	Small networks
Class D	224 ~ 239	Multicasting
Class E	240 ~ 255	Experimental

❓ Using above ranges, you can determine the class of an address from its 1st octet value

Classful IP Address Scheme: Classes (Cont.)

▶ E.g. an address having its first octet value:

- 120 -> is a Class A address
- 155 -> is a Class B address
- 220 -> is a Class C address

➤ Governing rules:

1. Leading bit(s) of the first octet is significant bit(s) which cannot be used to accomodate network addresses (i.e. each address starts with them)
 - Class A significant bit: 0
 - Class B significant bits: 10
 - Class C significant nits: 110
2. Cannot be designated by all zeroes or all ones
 - Applies to both Network and Host parts

Are You the Host or the Network?

- ▶ The 32 bits of the IP address are divided into Network & Host portions, with the octets assigned as a part of one or the other

Classful IP Address Scheme IP Address Structure (per class):				
Class	Octet 1	Octet 2	Octet 3	Octet 4
Class A	Network	Host	Host	Host
Class B	Network	Network	Host	Host
Class C	Network	Network	Network	Host
Class D	Multicast – No network/host concept			
Class E	Reserved for experimental research			

i Each Network is assigned a **network address** & every device or interface (such as a router port) on the network is assigned a **host address**

Class A IP Addresses

- ▶ The leading bit (1st Octet) which is always 0, including the remaining 7 bits are used to designate the Network as Class A
- ▶ There are 3 host octets:
 2^{24} or 16,777,216 hosts
- ▶ The formula goes as:
 $2^n = \text{Number of available/usable hosts}$
 - Note: The first address is always a network address and last address is a broadcast address i.e., $2^n - 2$
- ▶ You can use the same formula to determine the number of Networks in any address class
 - E.g. a Class A address uses 7 bits to designate the network, so: $2^7 = 128$ or there can be 128 Class A Networks
 - The significant bit is never counted

Class B IP Addresses

- ▶ The leading 2 bits (1st Octet) which are always 10, including the remaining 14 bits are used to designate the Network as Class B
 - ▶ This leaves 16 bits (two octets) to designate the Hosts
- ▶ So how many Class B Networks can there be?
 - ▶ Using our formula, (2^{14}) , there can be 16,384 Class B Networks & each Network can have (2^{16}) Hosts, or 65,536 Hosts

Class C IP Addresses

- ▶ The leading 3 bits of all class C addresses are set to 110, leaving 21 bits for the Network address, which means there can be 2,097,152 (2^{21}) Class C Networks
- ▶ 256 (2^8) Hosts per Network

→ For no. Of Networks:

'Identify bits' are not counted

→ For no. of Networks or Hosts, all zeros (broadcast) or all ones (default route) bits are not counted

→ For no. usable Hosts:

Network, Default Gateway and Broadcast addresses are not counted

In Short

Class	Address Range	Identify Bits (binary value)	Bits in Network part	Number of Networks	Bits in Host part	Number of Hosts/ Network
A	0 ~ 127	1 (0)	7	126	24	16,777,214
B	128 ~ 191	2 (10)	14	16,382	16	65,534
C	192 ~ 223	3 (110)	21	2,097,150	8	254

Special Addresses

- ▶ A few addresses are set aside for specific purposes e.g., in Cisco 'realm'

Network Address	Host Address	Description	Example
0's	0's	Default Cisco Route	0.0.0.0
0's	Host Address	Local Network Hosts	0.0.0.115
1's	1's	Broadcast to Local Network	255.255.255.255
Network Address	1's	Broadcast to Network Address	192.21.12.255
127	Anything	Loopback Testing	127.0.0.1

i For full list:

<https://www.iana.org/assignments/iana-ipv4-special-registry/iana-ipv4-special-registry.xhtml>

Special Addresses (Cont.)

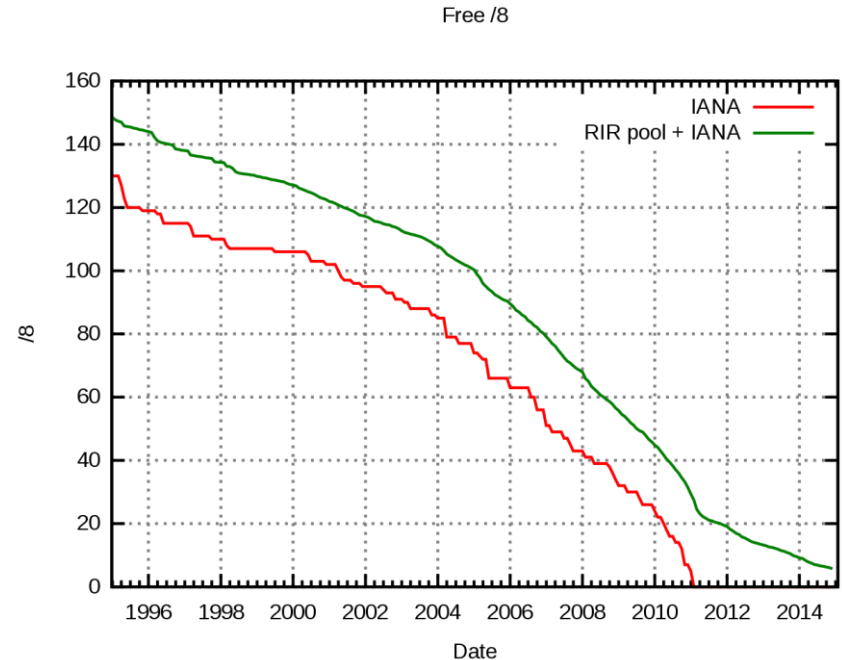
- ▶ **Within** each address class is a set of addresses that are set aside **for use in Local networks**, sitting behind a firewall or NAT (Network Address Translation) device, or Networks not connected to the Internet
- ▶ A list of these addresses for each IP address class:

IP Class	Address Range (Local)
Class A	10.0.0.0 ~ 10.255.255.255
Class B	172.16.0.0 ~ 172.31.255.255
Class C	192.168.0.0 ~ 192.168.255.255

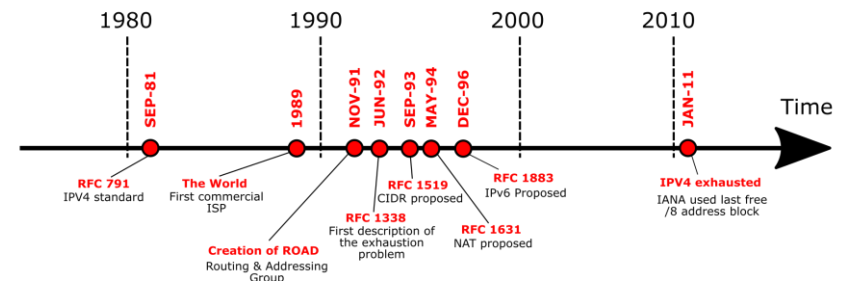
Limitations to Classful Addressing

- ▶ Limited IPv4 address space
 $2^{32} = 4,294,967,296$ addresses
- ▶ Class boundaries restricts efficient allocation of address space
- ▶ Lack of address class to support medium size company
 - Class B: 65534 hosts/network
→ *too large!*
 - Class C: 254 hosts/network,
→ *too small!*

New direction:
IPv6 (128 bits or 16 bytes/octets)



Source: Mro. "Exhaustion of IPv4 addresses since 1995." CC BY-SA 4.0. Image based on data from IANA and Potaroo.net



Source: Mro, IPv4 exhaustion timeline, Wikipedia, CC BY-SA 4.0.

Subnet Mask

- ▶ As we've seen, an IP address has 2 parts:
 - Network ID
 - Host ID
- ▶ Frequently, the Network & Host portions of the address need to be separately extracted
 - E.g. In classful IP scheme, if you know the address class, it's easy to separate the 2 portions.
- ▶ **To tackle IPv4 shortage / avoid wastage:**
 - ❶ **Idea:** borrow additional bits from network or host part i.e. Network or Host (VLSM – Variable length subnet mask)

Subnet Mask (Cont.)

- ▶ Determines whether an IP address exists on the local network or whether it must be routed outside the local network
 - ▶ It is applied to a message's destination address to extract the network address
 - ▶ If the extracted network address matches the local network ID, the destination is located on the local network
- ▶ Following are the **default subnet masks in IP Address Classful scheme** for Class A, B and C addresses:

Address Class	Subnet Mask
Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

A Trial Separation (good to know)

- ▶ Using subnet mask to determine **Network ID** of the destination network:
 1. If a destination IP address is 206.175.162.21, we know that it is a Class C address & that its binary equivalent is:
11001110.10101111.10100010.00010101
 2. We also know that the default standard Class C subnet mask is: 255.255.255.0 and that its binary equivalent is:
11111111.11111111.11111111.00000000

A Trial Separation (Cont.)

3. When these two binary numbers (the IP address & the subnet mask) are combined using Boolean Algebra (AND operation), the result is the **Network ID** of the destination network:

Destination IP

Decimal IP Address

Binary

206.175.162.21

11001110.10101111.10100010.00010101

Default Class C Subnet mask

255.255.255.0

11111111.11111111.11111111.00000000

Result (ANDed)

11001110.10101111.10100010.00000000

Network ID of the destination network

i If the result is the **IP address of the network** matches with one of the internal networks, then the packet belongs to the local network)

Exercise: Configuration of an interface

- ▶ From the configuration terminal
- ▶ Interface <name> <place> / <number>

```
epita(config)#interface fastEthernet 0/0  
epita(config-if)#ip address 192.168.1.1 255.255.255.0  
epita(config-if)#no shutdown
```

Note: Run the command “no shutdown” to activate the interface

- ▶ Interface loopback configuration

```
epita(config)#interface loopback 0  
epita(config-if)#ip address 192.168.0.1 255.255.255.0  
epita(config-if)#no shutdown
```

Exercise: Configuration of an interface (cont.)

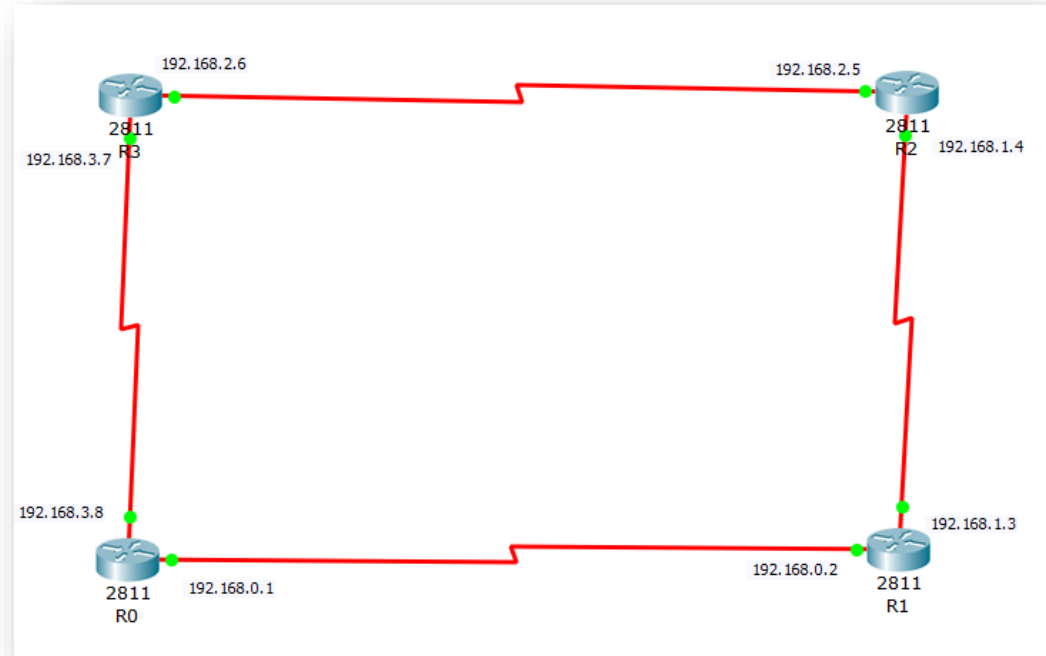
- ▶ The show interface command allows you to check the interface status

```
epita#show interfaces Serial 0/0/0  
Serial0/0/0 is up, line protocol is up
```

- ▶ An interface can be “administratively down”
 - “down” here a cable or material problem
- ▶ Line protocol is “down” is a configuration problem
- ▶ The “show ip interface brief” give you access to a simple plan of interface

Exercise 4: Practical work

- ▶ Using network simulation program:
 - ▶ Implement the schematic diagram (the Router names and IP addresses must match those specified in the network topology diagram)



*Save cisco packet tracer file with your:
First_Last-name_Ex-3-4'*

Deadline: See 'Teams' Assignment section

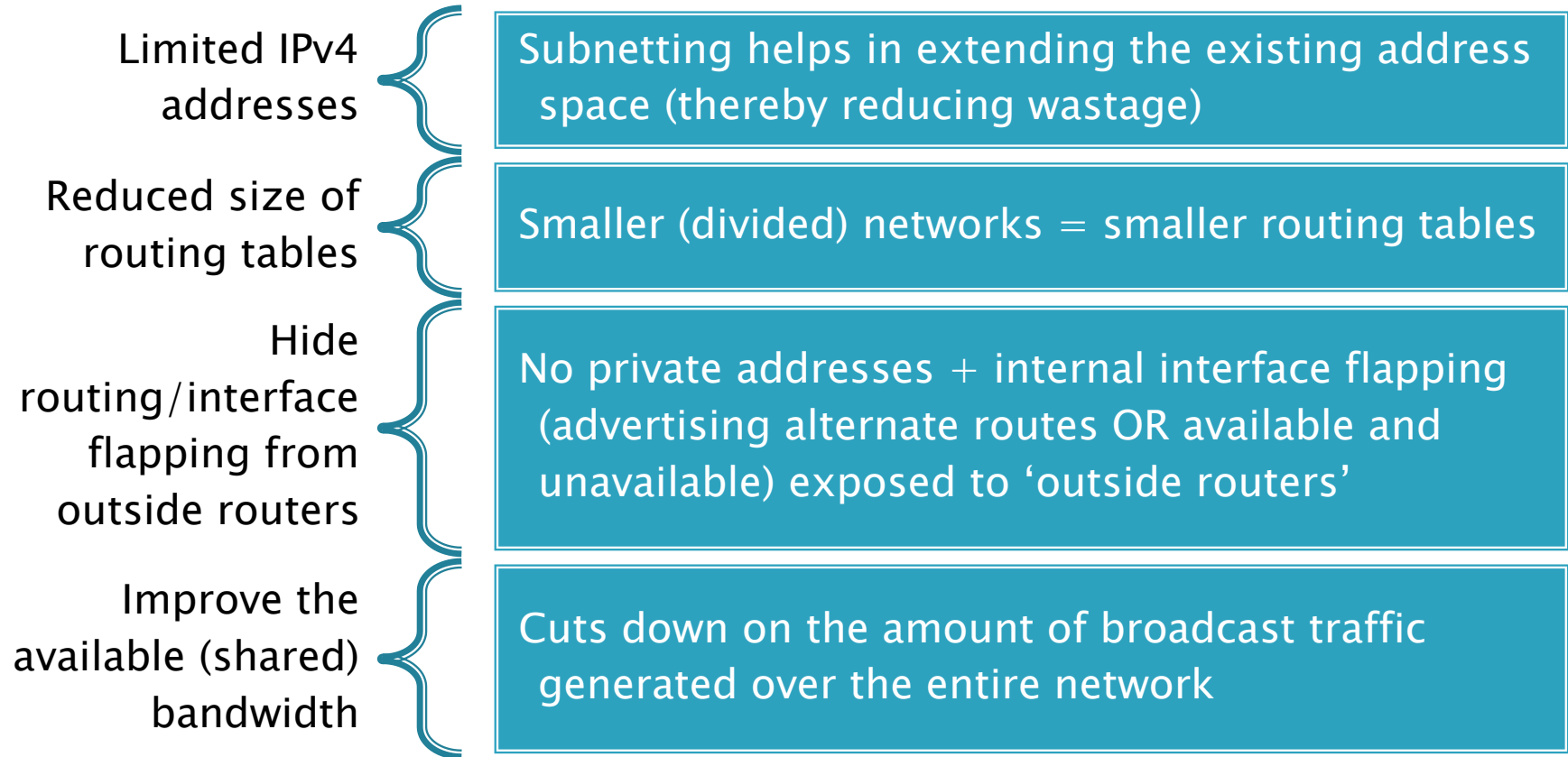
Lecture 2 Outline

- ▶ IP address
 - Allocation & Routing
 - Schemes and classes
 - Limitations to Classful Addressing
 - Subnet Mask & VLSM
 - Class exercise 4
- ▶ Subnetting
 - Subnetting, Subnet & Subnet Mask
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Subnetting, Subnet & Subnet Mask

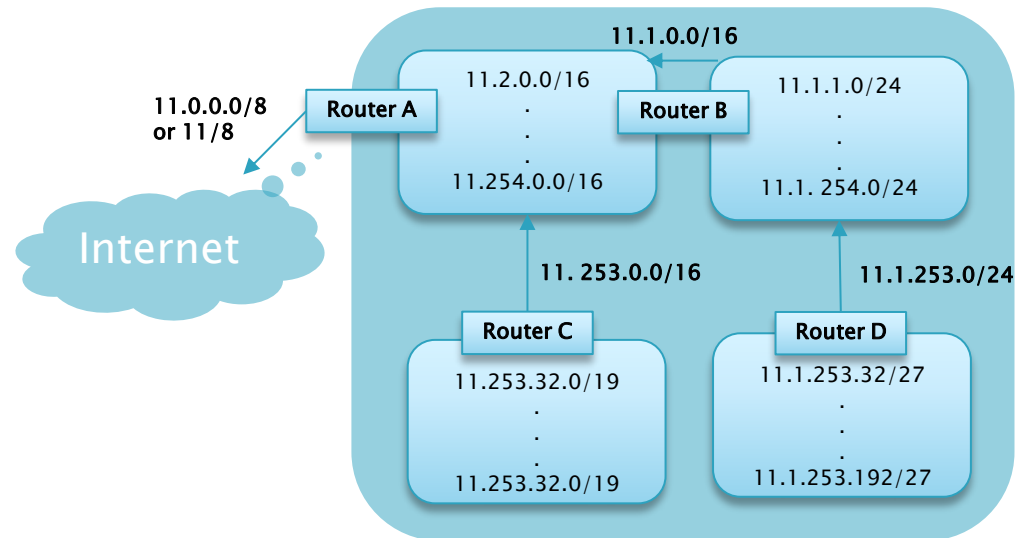
- ▶ Subnetting is the process of dividing a network & its IP addresses into segments, each of which is called a subnetwork or subnet
 - The subnet mask is a 32-bit number that the router uses to cover up the network address, to show which bits are being used for identifying a subnet
- ▶ Subnetting, a subnet & a subnet mask are all different
 - In fact, the 1st creates the 2nd & is identified by the 3rd

Why Subnetting is important?



Subnetting (a single advertisement)

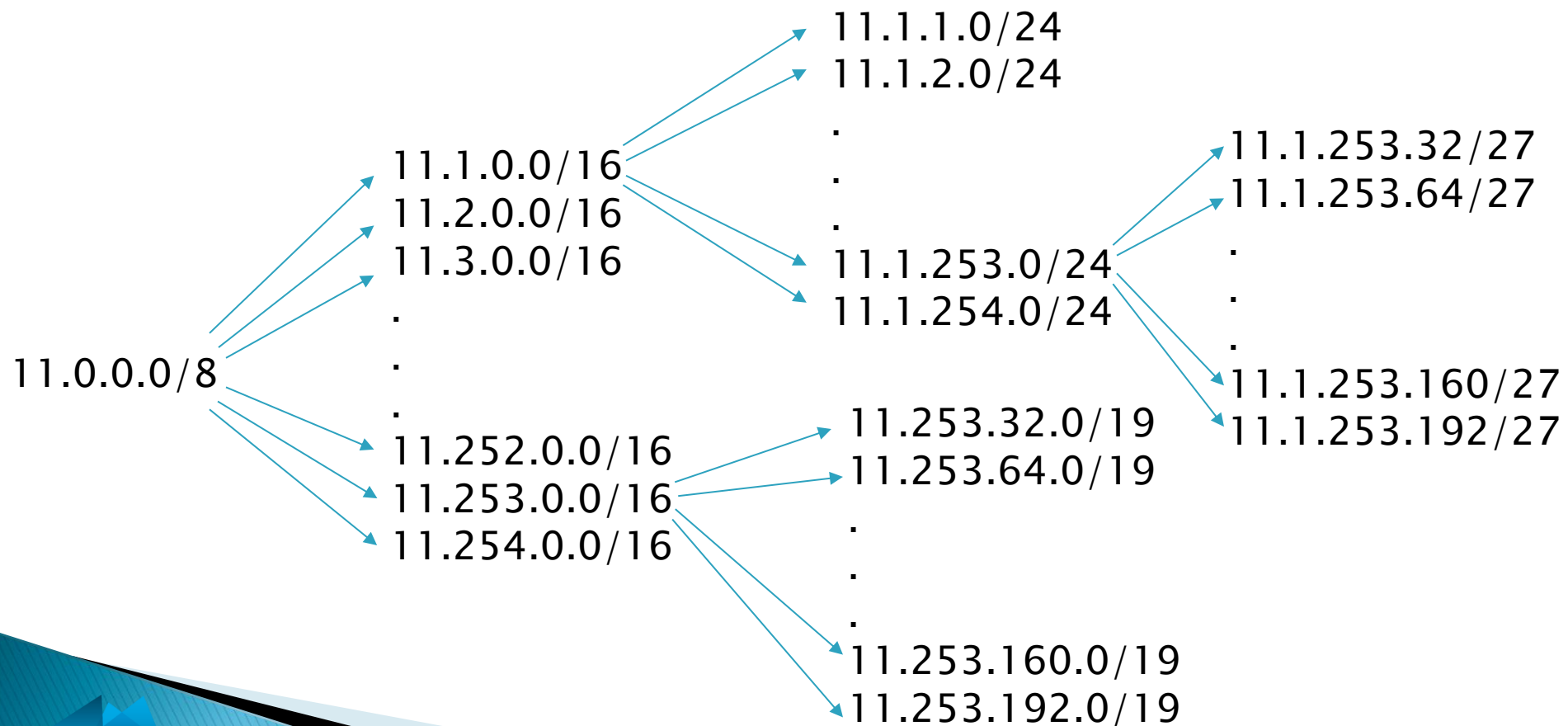
- ▶ A subnet summarizes all its lower level hierarchies into a single advertisement (or hides internal subnets from outside network)
- ▶ Allows the expansion of both Local Networks & the Internet in today's world
 - A must-know for network administrators



i A single advertisement to the outside world!

Subnet(s) (recursive division)

- ▶ Helps to reduce routing table size (Route Aggregation)
 - Can be further divided recursively while reducing the IP addresses wastage



VLSM and CIDR

- ▶ Variable Length Subnet Mask (VLSM):
 - Carrying bits from host portion (extended-network-prefix)
 - OR vice versa; depends on the requirements
 - Allow more efficient use of network addresses
 - Represented as e.g., 255.255.255.128
- ▶ Classless Inter-Domain Routing (CIDR):
 - Gets translated into subnet mask
 - Represented as e.g., /25
 - Eliminates the class concept

Borrowing Bits to Grow a Subnet

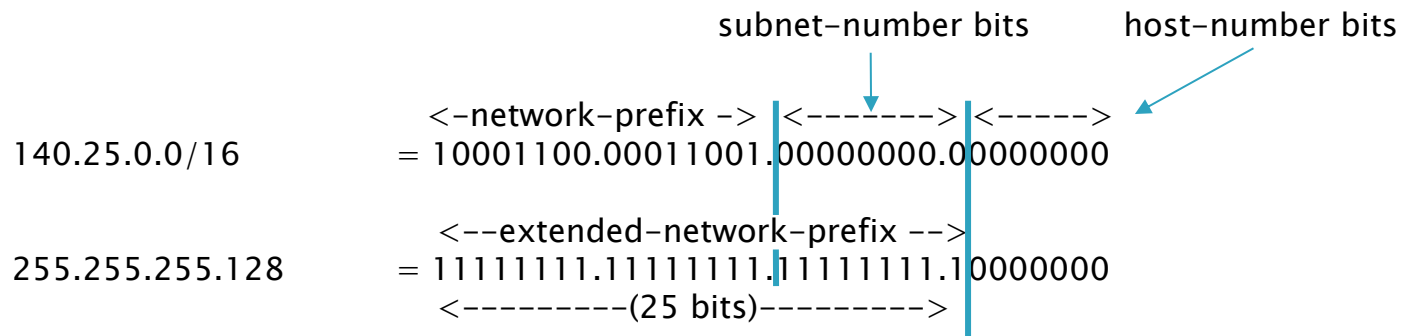
- ▶ The key concept in subnetting is borrowing bits from the host portion of the network to create a subnetwork (or vice versa, based on the requirements)
 - ▶ Rules govern this borrowing, ensures that some bits are left for a Host ID
 1. 2 bits remain available to use for the Host ID (allowing at least 3 Host addresses)
 2. Subnet bits cannot be all 1s or 0s at the same time
- i** To split up a class C address into variable length subnets, **always start with the largest subnet first** and then work your way down e.g. /26 should be first, then /27s, /29's ...

Subnetting (design example)

- ▶ An organization has been assigned with the network number **140.25.0.0/16** and it needs to create set of subnets that supports up to **60** hosts on each subnet (with enough room for expansion)

1: Defining the Subnet Mask (or Extended-Network-Prefix)

- $2^6 = 64$, not enough room for expansion
- $2^7 = 128 \rightarrow$ *Network part: 25 bits; Host part: 7 bits*



Base Net: 10001100.00011001.00000000.00000000 = 140.25.0.0/16

2: OPTIONAL: Defining Subnet (SN) addresses

Took 9 bits from the host part $\rightarrow 2^9 = 512$ (total)

SN #0: 10001100.00011001.00000000.00000000 = 140.25.0.0/25

SN #1: 10001100.00011001.00000000.10000000 = 140.25.0.128/25

...

SN #256: 10001100.00011001.11111111.10000000 = 140.25.255.128/25

3: Defining Hosts (+ broadcast) Addresses for Each Subnet

SN #0: 10001100.00011001.00000000.00000000 = 140.25.0.0/25

Host #1: 10001100.00011001.00000000.0 0000001 = 140.25.0.1/25

Host #2: 10001100.00011001.00000000.0 0000010 = 140.25.0.2/25

...

Host #126: 10001100.00011001.00000000.0 1111110 = 140.25.0.126/25

i 7 Host bits $\rightarrow 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 127$ (Host range)
Usable Hosts: $127 - 2 = 125$

Default gateway

SN #0: Broadcast (all 1's in host part)

10001100.00011001.00000000.0 1111111 = 140.25.0.127

Subnet Design Considerations

1. How many total subnets does the organization need today?
2. How many total subnets will the organization need in the future?
3. How many hosts are there on the organization's largest subnet today?
4. How many hosts will there be on the organization's largest subnet in the future?
5. ...

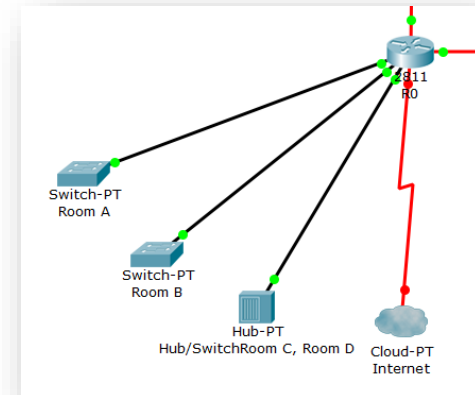
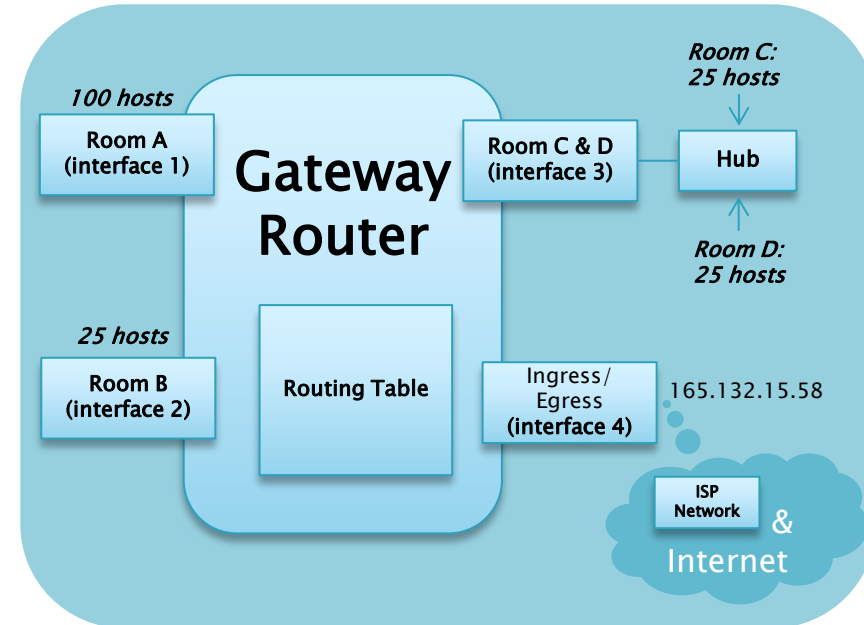
Exercise 5: Practical work

- ▶ Use your existing cisco packet tracer file (from last class exercise):

1. Subnet the given IP address: 165.132.9.0/24
 - Under the requirements as shown in the diagram:
2. Implement subnetted networks on R0 (except the Cloud-PT interface, which acts as an (unconfigured) all-pass (or a default/internet) route for demonstration purposes only)

Save cisco packet tracer file with your 'First_Last-name_Ex-3-5'

Deadline: See 'Teams' Assignment section



Subnet for Room A

Exercise 5: Subnetting

Room A (100 PCs):	100	IP addresses
Network Destination:	1	IP address $\leftarrow$ Lowest IP address in subnet
Broadcast IP:	1	IP address $\leftarrow$ Highest IP address in subnet
Gateway Interface 1:	1	IP address $\leftarrow$ Second highest IP address in subnet
IP Addresses Required	103	$\leq 128 = 2^7$

→ Subnet mask 11111111.11111111.11111111.10000000 = 255.255.255.128
7 bits

IP address for Room A: 165.132.9.★

★ { 0	→ Network Destination
1-100	→ IP Addresses for 100 PCs
★ { <u>101-125</u>	→ IP Addresses not used (25 reserved for future use)
126	→ Gateway Interface 1 IP address
127	→ Broadcast IP Address

We needed $103 \leq 128 = 2^7 = \text{Subnet Size}$

Subnet mask: $255.255.255.128 \leftarrow (256 - 2^7) = 128$

CIDR (Classless Inter-Domain Routing)

- Network Destination / Subnet Mask 1's (32 – 7 = 25)
- CIDR Network ID: 165.132.9.0/25

Subnet for Room B

Exercise 5: Subnetting (Cont.)

Room B (25 PCs):	25	IP addresses
Network Destination:	1	IP address $\leftarrow$ Lowest IP address in subnet
Broadcast IP:	1	IP address $\leftarrow$ Highest IP address in subnet
Gateway Interface 2:	1	IP address $\leftarrow$ Second highest IP address in subnet
IP Addresses Required	28	$\leq 32 = 2^5$

→ Subnet mask 11111111.11111111.11111111.11110000 = 255.255.255.224
5 bits

IP address for Room B: 165.132.9.★

★ {	128	→ Network Destination
	129–153	→ IP Addresses for 100 PCs
	<u>154–157</u>	→ IP Addresses not used (4 reserved for future use)
	158	→ Gateway Interface 1 IP address
	159	→ Broadcast IP Address

We needed $28 \leq 32 = 2^5 = \text{Subnet Size}$

Subnet mask: $255.255.255.224 \leftarrow (256 - 2^5) = 224$

CIDR (Classless Inter-Domain Routing)

- Network Destination / Subnet Mask 1's ($32 - 5 = 27$)
- CIDR Network ID: 165.132.9.128/27

Subnet for Room C & D

Exercise 5: Subnetting (Cont.)

Room C+D (50 PCs):	50	IP addresses
Network Destination:	1	IP address $\leftarrow$ Lowest IP address in subnet
Broadcast IP:	1	IP address $\leftarrow$ Highest IP address in subnet
Gateway Interface 3:	1	IP address $\leftarrow$ Second highest IP address in subnet
IP Addresses Required	53	$\leq 64 = 2^6$

→ Subnet mask 11111111.11111111.11111111.11000000 = 255.255.255.192
6 bits

IP address for Room C & D: 165.132.9.★

★	160	→ Network Destination
	161–210	→ IP Addresses for 50 PCs
	<u>211–221</u>	→ IP Addresses not used (11 reserved for future use)
	222	→ Gateway Interface 1 IP address
	223	→ Broadcast IP Address

We needed $53 \leq 64 = 2^6 = \text{Subnet Size}$

Subnet mask: $255.255.255.192 \leftarrow (256 - 2^6) = 192$

CIDR (Classless Inter-Domain Routing)

- Network Destination / Subnet Mask 1's ($32 - 5 = 27$)
- CIDR Network ID: 165.132.9.160/26

Exercise 5: Subnetting (a common mistake)

- ▶ Subnet for Room A → 165.132.9.0/25
Usable Host IP Range: 165.132.9.1 – 165.132.9.126
 - ▶ Subnet for Room B → 165.132.9.128/27
Usable Host IP Range: 165.132.9.129 – 165.132.9.158
 - ▶ Subnet for Room C & D → 165.132.9.160/26
Usable Host IP Range: 165.132.9.129 – 165.132.9.190
- IP range conflict!

i What should be done to correct this?
→ Check Slide no. 31 info-box

- ▶ Subnet for Room C & D → 165.132.9.128/26
Usable Host IP Range: 165.132.9.129 – 165.132.9.190
- ▶ Subnet for Room B → 165.132.9.192/27
Usable Host IP Range: 165.132.9.193 – 165.132.9.222

Routing: IP Routing Table

- ▶ Using the IP addresses from **Exercise 5**, following Routing Table will be maintained at the Gateway router:

Room	Destination (Network Destination)	Subnet Mask (or Netmask)	Gateway (GW)	Interface (GW Port)	Metric
A	165.132.9.0	255.255.255.128	165.132.9.126	1	10
C, D	165.132.9.128	255.255.255.192	165.132.9.190	3	10
B	165.132.9.192	255.255.255.224	165.132.9.222	2	1
Internet (Uplink)	165.132.15.56	255.255.255.252	165.132.15.58	4	10
Default Route	0.0.0.0	0.0.0.0	165.132.15.58	4	10

Metric indicates the associated cost (efficiency) of the route – lower is more preferred

- ▶ The last row has a Subnet Mask of 0.0.0.0 where all 32 bits of the Subnet Mask are 0s which makes it 'all pass' Subnet Mask
 - Therefore, the Internet row of the routing table can be combined into the last row of "Default Route"

Room	Destination (Network Destination)	Subnet Mask (or Netmask)	Gateway (GW)	Interface (GW Port)	Metric
A	165.132.9.0	255.255.255.128	165.132.9.126	1	10
C, D	165.132.9.128	255.255.255.192	165.132.9.190	3	10
B	165.132.9.192	255.255.255.224	165.132.9.222	2	1
Internet	0.0.0.0	0.0.0.0	165.132.15.58	4	10

→ *"Default Route" row and "Internet (Uplink)" row are combined*

- ▶ When an IP packet enters the Gateway/Router the routing table is used to determine where to send this packet out
- ▶ Upon receiving the IP packet,
 - The 1st row is checked first
 - Then going down, each row is sequentially checked in the routing table:



Room	Destination (Network Destination)	Subnet Mask (or Netmask)	Gateway (GW)	Interface (GW Port)	Metric
A	165.132.9.0	255.255.255.128	165.132.9.126	1	10
C, D	165.132.9.128	255.255.255.192	165.132.9.190	3	10
B	165.132.9.192	255.255.255.224	165.132.9.222	2	1
Internet	0.0.0.0	0.0.0.0	165.132.15.58	4	10

Verifying an IP Address

- ▶ IP addresses connectivity can be checked using PING (ICMP), Trace, Telnet, ...
- ▶ It is important that you know that PING is used to verify IP address connections to the Network Layer & that Telnet is used to verify network IP address connections to the Application Layer

Verifying with Traceroute (tracert)

- ▶ The Traceroute (GNU/Linux) or Tracert (Windows) command is used to show the complete route from a source to a destination
 - It sends out probe packets one at a time to each router/hop/switch/... in the path between the source & the destination IP address entered
 - Displays the round-trip time for each packet sent to each upstream router

i It is used to determine where a breakdown in a route may be occurring

Trace command response codes	
Response	Meaning
*	Timed out
!H	Router received packet but did not forward it (usually due to an access list)
N	Network unreachable
P	Protocol unreachable
U	Port unreachable

Lecture 2 Outline

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- ▶ Subnetting
 - Subnetting, Subnet & Subnet Mask
 - VLSM & CIDR
 - Calculating Subnets
 - Class exercise 5
 - Routing Table run-down
- ▶ **ARP**
 - **Frame & working**
 - **Tables**
 - **Class exercise 6**

ARP Protocol

- ▶ Address Resolution Protocol (ARP) protocol allows getting a physical address (MAC) from a logical address (IP) and vice versa
- ▶ Devices store MAC–IP bindings:
 - Switch maintains a Forwarding Table
 - Router maintains a MAC Table
 - End–devices/hosts maintains ARP cache
- ▶ This allow hosts, connected to the same LAN, to communicate without passing the gateway (the destination MAC is already present in ARP cache)

How ARP works?

- ▶ Scenario: Host A wish to talk to host B, ARP cache is empty:
 1. Host 'A' sends a broadcast request in order to get the host 'B' (192.168.0.2) MAC address
 2. Host 'B' receives the request, recognizes its IP address and then sends a response containing its MAC address to host 'A'
 3. Host 'A' updates its ARP cache and can now communicate with host 'B'

Host A (192.168.0.1)



"Who has 192.168.0.2?"
Destination: FF:FF:FF:FF:FF:FF (broadcast to all hosts on the local network)

Local Area Network

Host B (192.168.0.2)



Reply: "192.168.0.2 is at 00:50:fc:70:10:4d"
Destination: Host A

ARP Tables

- ▶ Two entries types:
 - Dynamic: That's the default way ARP work and correspondences are temporary stored (depending on the OS)
 - Static: In this case, the correspondence is manually registered without any duration limit
- ▶ ARP Table (router)

```
R0#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  165.132.9.126    -         0060.70AE.2B92  ARPA   FastEthernet0/0
Internet  165.132.9.190    -         0060.709B.7D49  ARPA   FastEthernet6/0
Internet  165.132.9.222    -         00E0.A339.846C  ARPA   FastEthernet1/0
```

- ▶ Forwarding Table (switch)

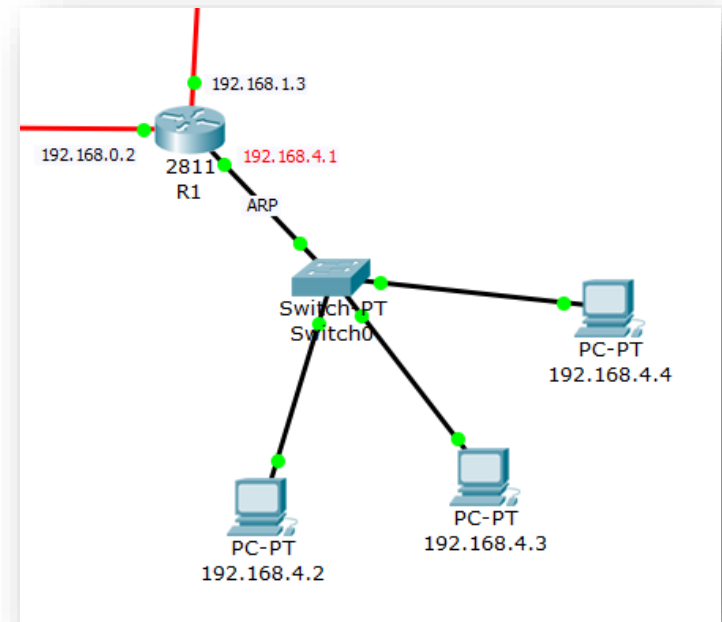
```
Switch#show mac-address-table dynamic
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0060.70ae.2b92   DYNAMIC     Fa0/1
```

- ▶ ARP cache (host)

```
C:\>arp -a
No ARP Entries Found
C:\>arp -a
Internet Address      Physical Address      Type
192.168.4.1           0001.6407.7820        dynamic
```


Exercise 6: Practical work

- ▶ Use your existing cisco packet tracer file (from last class exercise):
 1. Attach PC1, PC2 and PC3 to R1
 2. Configure them using 192.168.4.* IPv4 addresses
 3. Check PC1 arp table (should be empty)
 4. Initiate PING from PC1 to PC3 to set-up the ARP – and simulate the connection
 5. Verify ARP setup (using arp -a)



Save cisco packet tracer file with your 'First_Last-name_Ex-3-6'

Deadline: See 'Teams' Assignment section

Lecture 2 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**